# Brianna Hinds

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## Education

**Houston Christian University** Houston, TX
*Bachelor of Science in Computer Science, Minor in Data Analytics* *August 2022 – May 2026*

- **Extracurricular**: Hurdler on the HCU Track Team, President/Founder of SOURCE
- **Relevant Coursework**: Data Analytics, Artificial Intelligence, Data Structures, Database Management Systems, Theory of Computing

## Professional Work Experience

**ML Engineer** June 2025 – Present
*Sysco* *Houston, TX*

- Engineered a cloud-native, 4-tier machine learning pipeline to automate ERP-to-MDM item mapping, accelerating supply chain data processing times by 80%.
- Designed a hierarchical matching engine combining deterministic rules, fuzzy string logic (RapidFuzz), and semantic K-Nearest Neighbor searches (SentenceTransformers, FAISS) to resolve complex vendor data.
- Refactored the monolithic AWS SageMaker model into a scalable, serverless MLOps architecture using Kubeflow Pipelines on Google Cloud Platform (Vertex AI).

**Data Science Researcher** August 2024 – April 2025
*Purdue University* *West Lafayette, IN, Remote*

- Analyzed Houston Fire Department's operational records to identify critical bottlenecks and develop data-driven strategies for optimizing resource allocation.
- Engineered comprehensive data pipelines using Python and R to clean, process, and extract actionable insights from complex municipal datasets.
- Designed and deployed interactive visualization tools to clearly communicate performance metrics and predictive response times to key city stakeholders.

## Projects

**Racr: F1 Strategy Platform** | *Python, XGBoost, Streamlit Plotly, Pandas, FastF1* January 2025 – Present

- Built an end-to-end race strategy simulation system predicting lap-by-lap and total race time using an XGBoost regression model.
- Engineered race-aware features (tire degradation, stints, pit timing, fuel effects, weather, in/out laps, track characteristics).
- Structured the project using MLOps-aligned practices (feature pipelines, model versioning, cached inference, modular architecture).
- Expanding feature engineering and model validation to improve generalization across tracks and race conditions.
- Achieved high predictive performance ($R^2 = 0.998$) while prioritizing strategy sensitivity and interpretability over pure metric optimization.

**Surge** | *Python, LangGraph, XGBoost, OpenAI API, Pandas, nmap* June 2025 – May 2026

- Design and implement an intelligent multi-agent cybersecurity framework leveraging large language models (LLMs) to autonomously perform network reconnaissance, vulnerability detection, and executive-level reporting.
- Develop a Vulnerability Agent that integrates the CVE.CIRCL API with a custom XGBoost-based CVSS scoring model, achieving a 99% variance explanation in automated risk classification and prioritization.
- Engineer adaptive data normalization pipelines using Pydantic schemas and LangChain tool binding, enabling seamless interoperability between agents and reducing post-scan formatting errors by 80%.

## Technical Skills

**Languages**: Python, SQL, R, Java, Arduino/C++, HTML, CSS, JavaScript, C
**Frameworks/Technologies**: Object-Oriented Programming, Machine Learning Models, Neural Networks, PyTorch
**Skills**: Time Management, Communication, Teamwork, Leadership, Microsoft Excel, Data Cleansing, Git, GitHub